

Two-level Galerkin–Lagrange multipliers method for the stationary Navier–Stokes equations[☆]

Liping Zhu^{a,b}, Yinnian He^{a,*}

^a Faculty of Science, State Key Laboratory of Multiphase Flow in Power Engineering, Xi'an Jiaotong University, Xi'an 710049, People's Republic of China

^b College of Science, Xi'an University of Architecture and Technology, Xi'an, 710054, People's Republic of China

ARTICLE INFO

Article history:

Received 21 June 2007

Received in revised form 1 April 2008

Keywords:

Navier–Stokes equations

Galerkin–Lagrange multiplier

Error estimate

ABSTRACT

In this paper, we combine the Galerkin–Lagrange multiplier (GLM) method with the two-level method to solve the stationary Navier–Stokes equations in order to avoid the time-consuming process and the construction of zero-divergence elements. Different quadrilateral partitions are used for approximating the velocity and the pressure. Then some error estimates are obtained and some numerical results of the GLM method and the two-level GLM method are given. The results show that the two-level method based on the GLM method is more efficient than the GLM method under the convergence rate of same order.

© 2009 Elsevier B.V. All rights reserved.

1. Introduction

In this paper, we construct and analyze the Galerkin–Lagrange multiplier approximation $(u^h, p^{\hat{h}})$ to the solution pair (u, p) of the steady Navier–Stokes equations. The emphasis is put on seeking u^h in a finite-dimensional approximating space with elements that do not necessarily satisfy the zero-divergence condition. One such method, motivated by the Lagrange multiplier method of Babuska [1], was applied in [2] to the stationary Stokes equations and in [3] to the stationary Navier–Stokes equations.

In practice, the numerical solution of the nonlinear system of equations arising in the discretization of the Navier–Stokes equations may be very time consuming. Two-level methods aim to compute a discrete approximation of the solution of the nonlinear partial differential equation with less computational work, and to preserve the optimal order of convergence, e.g., see Xu [4,5] for the steady semi-linear elliptic equations and nonlinear elliptic equations, Layton [6], Layton and Lenferink [7] and Layton and Tobiska [8] and Girault and Lions [9] and He et al. [10,11] for the steady Navier–Stokes equations. As for the nonstationary Navier–Stokes equations, the two-level finite element semi-discretization scheme has been studied in [9], and the full discretization of the two-level finite element method in space variable x and the one-level backward Euler scheme in time variable t have been discussed in [12], and the full discretization of the two-level finite element method in the space-time variables x and t has been studied in [13–15], where the numerical test is provided in [15]. Recently, some multi-level strategy has been studied for the nonstationary Navier–Stokes equations in [16–18].

The method we study in this paper is combining the Galerkin–Lagrange multiplier method with the two-level discretization for solving the two-dimensional steady Navier–Stokes problem under the assumptions of the uniqueness condition.

In our approach, we seek u^h , the approximation to u , in a finite-dimensional subspace X_h of $H_0^1(\Omega)^2$, and $p^{\hat{h}}$, in a finite-dimensional subspace $M_{\hat{h}}$ of $H^1(\Omega)$; X_h and $M_{\hat{h}}$ corresponding to, in general, different quadrilateral partitions of Ω . If h and

[☆] This research is partly supported by the NSF of China (No. 10671154, No. 40730424) and the National Basic Research Program (No. 2005CB321703).

* Corresponding author.

E-mail address: heyn@mail.xjtu.edu.cn (Y. He).

$\widehat{h}$ are the discretization parameters for X_h and $M_{\widehat{h}}$, respectively, then most results in this paper hold provided $h/\widehat{h}$ is chosen sufficiently small.

The outline of the paper is as follows. In the next section we introduce the mathematical setting of the stationary Navier–Stokes equations. In Section 3 we recall the conclusions of the Galerkin–Lagrange multiplier method for the stationary Navier–Stokes equations. The optimal error estimate of the two-level Galerkin–Lagrange multiplier method is introduced in Section 4. Some numerical results are presented in Section 5, which show that our method is efficient.

2. Functional setting of the Navier–Stokes equations problem

Let Ω be a bounded domain in $\mathbb{R}^2$ assumed to have a Lipschitz continuous boundary $\partial\Omega$ and to satisfy a further condition stated in (2.2) below. We will consider the following Navier–Stokes problem: Find $u = (u_1, u_2)$ and p defined on Ω such that

$$\begin{cases} -\nu \Delta u + (u \cdot \nabla)u + \text{grad } p = f & \text{in } \Omega, \\ \text{div } u = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (2.1)$$

where u is the velocity, p is the pressure, f is the body forces per unit mass, and $\nu > 0$ is the dynamic viscosity.

For the mathematical setting of the problem, we introduce the following Hilbert spaces

$$X = H_0^1(\Omega)^2, \quad Y = L^2(\Omega)^2, \quad M = H^1(\Omega)/\mathbb{R}.$$

The spaces $L^2(\Omega)^m$, $m = 1, 2, 4$ are endowed with the L^2 -scalar product and L^2 -norm denoted by $(\cdot, \cdot)$ and $\|\cdot\|_0$. The spaces $H_0^1(\Omega)$ and X are equipped with their usual scalar product $(\nabla u, \nabla v)$ and norm $\|\nabla u\|_0$. For $(u, p) \in X \times L^2(\Omega)/\mathbb{R}$, we let

$$\|(u, p)\| = \{\|\nabla u\|^2 + \|p\|_{0/R}^2\}^{1/2},$$

where $\|p\|_{0/R} = \inf_{c \in \mathbb{R}} \|p + c\|_0$. Also, we denote by A the unbounded linear operator given by

$$Au = -\Delta u \quad \forall u \in D(A) = H^2(\Omega)^2 \cap X \quad (2.2)$$

and the bilinear operator $B(\cdot, \cdot)$ is defined by

$$B(u, v) = (u \cdot \nabla)v + \frac{1}{2}(\text{div } u)v \quad \forall u, v \in X.$$

Moreover, we define the continuous bilinear forms $a(\cdot, \cdot)$ and $d(\cdot, \cdot)$ on $X \times X$ and $X \times M$ respectively, by

$$a(u, v) = \nu(\nabla u, \nabla v) \quad \forall u, v \in X, \quad d(v, p) = (v, \nabla p) \quad \forall v \in X, p \in M$$

and a generalized bilinear form on $(X, M) \times (X, M)$, by

$$\mathcal{B}((u, p), (v, q)) = a(u, v) + d(u, q) + d(v, p) \quad \forall (u, p), (v, q) \in X \times M.$$

and a trilinear form on $X \times X \times X$ by

$$\begin{aligned} b(u, v, w) &= \langle B(u, v), w \rangle_{X', X} = ((u \cdot \nabla)v, w) + \frac{1}{2}((\text{div } u)v, w) \\ &= \frac{1}{2}((u \cdot \nabla)v, w) - \frac{1}{2}((u \cdot \nabla)w, v) \quad \forall u, v, w \in X. \end{aligned}$$

It is well known that $b(\cdot, \cdot, \cdot)$ satisfies the following properties (see [7,19]):

$$b(u, v, w) = -b(u, w, v) \quad \forall u, v, w \in X \quad (2.3)$$

$$\begin{aligned} |b(u, v, w)| &\leq c_0 \|u\|_0^{1/2} \|\nabla u\|_0^{1/2} \|\nabla v\|_0 \|w\|_0^{1/2} \|\nabla w\|_0^{1/2} \\ &\quad + c_0 \|\nabla u\|_0 \|v\|_0^{1/2} \|\nabla v\|_0^{1/2} \|w\|_0^{1/2} \|\nabla w\|_0^{1/2} \quad \forall u, v, w \in X, \end{aligned} \quad (2.4)$$

$$|b(u, v, w)| \leq C_b \|\nabla u\|_0 \|\nabla v\|_0 \|\nabla w\|_0 \quad \forall u, v, w \in X. \quad (2.5)$$

where c_0, C_b are positive constants depending on the domain Ω

Then the finite element Galerkin approximation scheme is to find $(u, p) \in X \times M$ such that

$$a(u, v) + d(v, p) + d(u, q) + b(u, u, v) = (f, v) \quad \forall (v, q) \in X \times M. \quad (2.6)$$

that is to find $(u, p) \in X \times M$ such that

$$\mathcal{B}((u, p), (v, q)) + b(u, u, v) = (f, v) \quad \forall (v, q) \in X \times M. \quad (2.7)$$

Let $h > 0$ be a real positive parameter. $\tau_h(\Omega)$, a partition of Ω for the velocity, is assumed to be regular in the usual sense (see [20]), i.e., for some $\sigma > 1$ and $0 < \omega < 1$,

$$h_K < \sigma \rho_K, \quad |\cos \theta_{iK}| \leq \omega, \quad i = 1, 2, 3, 4, \forall K \in \tau_h(\Omega),$$

where h_K is the diameter of element K , ρ_K is the diameter of the inscribed circle of element K , and θ_{iK} are the angles of K in the case of a quadrilateral partitioning. The mesh parameter h is given by $h = \max h_K$. $\tau_h(\Omega)$ is a partition of Ω for the pressure. Then

- (i) $X_h \subset X$ is a one-parameter family of finite-dimensional subspaces consisting of piecewise polynomial functions of degree 1 defined on $\tau_h(\Omega)$, and X_h possess the following approximation property:

$$\inf_{v_h \in X_h} \{ \|w - v_h\|_0 + h \|\nabla(w - v_h)\|_0 \} \leq ch^2 \|Aw\|_0 \quad \forall w \in D(A), \quad (2.8)$$

and the inverse inequality

$$\|v_h\|_{L^\infty} + \|\nabla v_h\|_0 \leq ch^{-1} \|v_h\|_0 \quad \forall v_h \in X_h. \quad (2.9)$$

- (ii) $M_{\hat{h}} \subset M$ is a one-parameter family of finite-dimensional subspaces consisting of piecewise polynomial functions of degree 1 defined on $\tau_{\hat{h}}(\Omega)$, and $M_{\hat{h}}$ possess the following approximation property:

$$\inf_{\chi \in M_{\hat{h}}} \{ \|p - \chi\|_0 + \hat{h} \|\nabla(p - \chi)\|_0 \} \leq c\hat{h} \|\nabla p\|_0 \quad \forall p \in M, \quad (2.10)$$

and

$$\|\nabla \chi\|_0 \leq c\hat{h}^{-1} \|\chi\|_0 \quad \forall \chi \in M_{\hat{h}}. \quad (2.11)$$

We also define

$$V_h = \{v \in X_h : (v, \text{grad } q) = 0, \forall q \in M_{\hat{h}}\}.$$

Proposition 2.1 ([3]).

- (i) There exists a constant $c = c(v, \Omega)$, such that

$$|\mathcal{B}((u, p), (v, q))| \leq c \| (u, p) \| \| (v, q) \| \quad \forall (u, p), (v, q) \in X \times M. \quad (2.12)$$

- (ii) There exists a constant β which is independent of $v, h, \hat{h}$, such that for some constant $c = c(\beta, v, \Omega, N)$ if $h/\hat{h} \leq \beta$, we have

$$\sup_{\substack{(v, q) \in X_h \times M_{\hat{h}} \\ \| (v, q) \| \leq 1}} |\mathcal{B}((u, p), (v, q))| \geq c \| (u, p) \| \quad \forall (u, p) \in X_h \times M_{\hat{h}}. \quad (2.13)$$

Proposition 2.2 ([3]). Let $(u, p) \in (X \cap H^2(\Omega)^2) \times M$. Then under the condition of Proposition 2.1, there exists a unique $(w, \pi) \in X_h \times M_{\hat{h}}$ satisfying

$$B((w, \pi), (v, q)) = B((u, p), (v, q)), \quad \forall (v, q) \in X_h \times M_{\hat{h}}. \quad (2.14)$$

Moreover, for some constant c independent of h and $\hat{h}$,

$$\|\nabla(u - w)\|_0 \leq ch(\|u\|_2 + \|p\|_{1/R}), \quad \|p - \pi\|_{0/R} \leq c\hat{h}(\|u\|_2 + \|p\|_{1/R}). \quad (2.15)$$

3. The finite element Galerkin–Lagrange multiplier method

For each $f \in Y$, we define the Galerkin approximation $(u_h, p_{\hat{h}})$ to (u, p) in the following way: Seek $(u_h, p_{\hat{h}}) \in X_h \times M_{\hat{h}}$, such that

$$a(u_h, v) + (u_h, \text{grad } q) + (v, \text{grad } p_{\hat{h}}) + b(u_h, u_h, v) = (f, v), \quad (3.1)$$

for all $(v, q) \in X_h \times M_{\hat{h}}$.

Theorem 3.1 ([3]). Under the conditions of Proposition 2.1, there exists a solution pair $(u_h, p_{\hat{h}})$ to (3.1); moreover, if $v^2 - C_b \|f\|_{-1} > 0$, then the solution is unique and $u_h \in V_h$ satisfying

$$\|\nabla u_h\|_0 \leq \frac{1}{v} \|f\|_{-1}.$$

Theorem 3.2 ([3]). Let $(u_h, p_{\hat{h}})$ be the solution of (3.1). Under the condition $\delta = v^2 - C_b \|f\|_{-1} > 0$ and that of Proposition 2.2, there exists a constant c independent of $h, \hat{h}$ such that

$$\|\nabla(u - u_h)\|_0 + \|p - p_{\hat{h}}\|_{0/R} \leq c\hat{h}(\|Au\|_0 + \|p\|_{1/R}),$$

and

$$\|u - u_h\|_0 \leq c\hat{h}^2(\|Au\|_0 + \|p\|_{1/R}),$$

where $(\|Au\|_0 + \|p\|_{1/R})$ is bounded if $f \in Y$.

4. The two-level Galerkin–Lagrange multiplier method

In this section, we choose $H, \hat{H}$ for coarse grids, $h, \hat{h}$ for fine grids and $H \gg h > 0, \hat{H} \gg \hat{h} > 0$, and construct associated conforming finite element spaces $X_H \times M_{\hat{H}}$ and $X_h \times M_{\hat{h}}$, where $X_H \times M_{\hat{H}} \subset X_h \times M_{\hat{h}}$. Then we will consider the following two-level Galerkin–Lagrange multiplier methods.

4.1. Simple two-level Galerkin–Lagrange multiplier method

Step I: Solve the Navier–Stokes problem on a coarse mesh, i.e., find $(u_H, p_{\hat{H}}) \in X_H \times M_{\hat{H}}$, such that

$$a(u_H, v) + d(v, p_{\hat{H}}) + d(u_H, q) + b(u_H, u_H, v) = (f, v) \quad \forall (v, q) \in X_H \times M_{\hat{H}}, \quad (4.1)$$

Step II: Solve the Stokes problem on a fine mesh, i.e., find $(u^h, p^{\hat{h}}) \in X_h \times M_{\hat{h}}$, such that

$$a(u^h, v) + d(v, p^{\hat{h}}) + d(u^h, q) + b(u_H, u_H, v) = (f, v) \quad \forall (v, q) \in X_h \times M_{\hat{h}}. \quad (4.2)$$

First we need to introduce the discrete analogue $A_h : X_h \rightarrow X_h$ of the Laplace operator given by

$$(A_h u_h, v_h) = (\nabla u_h, \nabla v_h) \quad \forall u_h, v_h \in X_h.$$

Next we give the approximation of $b(u, v, w)$:

Lemma 4.1. $\forall u_{h_1} \in X_{h_1}, v_h \in X_{h_2}, w \in Y$, the trilinear form b satisfies the following estimates:

$$|b(u_{h_1}, v_{h_2}, w)| \leq c \|\nabla u_{h_1}\|_0^{1/2} \|A_{h_1} u_{h_1}\|_0^{1/2} \|\nabla v_{h_2}\|_0 \|w\|_0, \quad (4.3)$$

$$|b(w, u_{h_1}, v_{h_2})| \leq c \|\nabla u_{h_1}\|_0^{1/2} \|A_{h_1} u_{h_1}\|_0^{1/2} \|\nabla v_{h_2}\|_0 \|w\|_0, \quad (4.4)$$

$$|b(w, u_{h_1}, v_{h_2})| + |b(u_{h_1}, v_{h_2}, w)| \leq c |\ln h|^{1/2} \|\nabla u_{h_1}\|_0 \|\nabla v_{h_2}\|_0 \|w\|_0, \quad (4.5)$$

where $h = \min\{h_1, h_2\}$.

Remark. (4.3)–(4.5) can be proven in a similar manner to that used in [21].

Theorem 4.2. Assume that u_h is the solution of (3.1), then there exists a constant $c = c(v, \Omega, f)$ such that

$$\|A_h u_h\|_0 \leq c.$$

Proof. From the definition of $A_h u_h$ and (2.8)–(2.9), we obtain

$$\|A_h u_h\|_0 = \sup_{v_h \in X_h} \frac{(\nabla u_h, \nabla v_h)}{\|\nabla v_h\|_0},$$

$$(\nabla u_h, \nabla v_h) = (\nabla u_h - \nabla u, \nabla v_h) + (Au, v_h) \leq (ch^{-1} \|\nabla(u - u_h)\|_0 + \|Au\|_0) \|\nabla v_h\|_0.$$

Combining these two relations and using Theorem 3.2, one finds $\|A_h u_h\|_0 \leq c$.

Theorem 4.3. Under the condition of Theorem 4.2, there exists a constant c independent of h and $\hat{h}$, such that the solution $(u^h, p^{\hat{h}})$ of (4.1)–(4.2) satisfies

$$\|\nabla(u - u^h)\|_0 + \|p - p^{\hat{h}}\|_{0/R} \leq c_1 \hat{h} + c_2 \hat{H}^2.$$

Proof. Suppose that (u_h, p_h) is the solution to problem (3.1), let $E = u_h - u^h$, $\eta = p_{\hat{h}} - p^{\hat{h}}$, then from (3.1) and (4.2) we obtain

$$a(E, v) + (E, \text{grad } q) + (v, \text{grad } \eta) + b(u_h - u_H, u_h, v) + b(u_H, u_h - u_H, v) = 0.$$

From Proposition 2.1(ii) and Lemma 4.1, we obtain

$$\begin{aligned} \|\nabla E\|_0 + \|\eta\|_{0/R} &\leq c \sup_{\substack{(v,q) \in X_h \times M_{\hat{h}} \\ \|(v,q)\| \leq 1}} |\mathcal{B}((E, \eta), (v, q))| \\ &= c \sup_{\substack{(v,q) \in X_h \times M_{\hat{h}} \\ \|(v,q)\| \leq 1}} (|b(u_h - u_H, u_h, v)| + |b(u_H, u_h - u_H, v)|) \\ &\leq c \sup_{\substack{(v,q) \in X_h \times M_{\hat{h}} \\ \|(v,q)\| \leq 1}} (C_b \|u_h - u_H\|_0 \|\nabla u_h\|_0^{\frac{1}{2}} \|A_h u_h\|_0^{\frac{1}{2}} \|\nabla v\|_0 + C_b \|u_h - u_H\|_0 \|\nabla u_H\|_0^{\frac{1}{2}} \|A_H u_H\|_0^{\frac{1}{2}} \|\nabla v\|_0) \\ &\leq C_b \|u_h - u_H\|_0 \|\nabla u_h\|_0^{\frac{1}{2}} \|A_h u_h\|_0^{\frac{1}{2}} + C_b \|u_h - u_H\|_0 \|\nabla u_H\|_0^{\frac{1}{2}} \|A_H u_H\|_0^{\frac{1}{2}}. \end{aligned}$$

Thanks to Theorem 3.2, we have

$$\|u_h - u_H\|_0 \leq \|u - u_H\|_0 + \|u - u_h\|_0 \leq c\hat{H}^2.$$

From Theorems 3.1 and 4.2, we know that $\|A_H u_H\|_0$, $\|A_h u_h\|_0$, $\|\nabla u_H\|_0$, $\|\nabla u_h\|_0$ are bounded. So we have

$$\|\nabla E\|_0 + \|\eta\|_{0/R} \leq c\hat{H}^2.$$

then by Theorem 3.1 and the above conclusions, we have

$$\begin{aligned} \|\nabla(u - u^h)\|_0 + \|p - p^{\hat{h}}\|_{0/R} &\leq \|\nabla(u - u_h)\|_0 + \|\nabla(u_h - u^h)\|_0 + \|p - p_{\hat{h}}\|_{0/R} + \|p_{\hat{h}} - p^{\hat{h}}\|_{0/R} \\ &\leq c_1 \hat{h} + c_2 \hat{H}^2. \end{aligned}$$

Hence, if we choose $\hat{H}$ such that $\hat{h} = o(\hat{H}^2)$, then the TGLM method is of the same order of convergence rate as that of the usual GLM method. However, our method is more simple, and this can also be obtained from the numerical results.

4.2. Newton two-level Galerkin–Lagrange multiplier method

Step I: Solve the Navier–Stokes problem on a coarse mesh, i.e., find $(u_H, p_{\hat{H}}) \in X_H \times M_{\hat{H}}$, such that

$$a(u_H, v) + d(v, p_{\hat{H}}) + d(u_H, q) + b(u_H, u_H, v) = (f, v), \quad (4.6)$$

for all $(v, q) \in X_H \times M_{\hat{H}}$.

Step II: Solve the general Stokes problem on a fine mesh, i.e., find $(u^h, p^{\hat{h}}) \in X_h \times M_{\hat{h}}$, such that

$$a(u^h, v) + d(v, p^{\hat{h}}) + d(u^h, q) + b(u_h, u_H, v) + b(u_H, u_h, v) = (f, v) + b(u_H, u_H, v), \quad (4.7)$$

for all $(v, q) \in X_h \times M_{\hat{h}}$. Next, we will study the convergence of the Newton two-level Galerkin–Lagrange multiplier solution $(u^h, p^{\hat{h}})$ to (u, p) in some norms.

Suppose that $(u, p) \in D(A) \times M$ is the solution to problem (2.1), we consider a dual linearized Navier–Stokes problem of finding $(z, s) \in X \times M$ satisfying

$$a(w, z) + b(u, w, z) + b(w, u^h, z) - (t, \text{div } z) + (s, \text{div } w) = (u - u^h, w), \quad (4.8)$$

for all $(w, t) \in X \times M$, then we have the following regularity results.

Theorem 4.4 ([22]). *There exists a unique solution $(z, s) \in D(A) \times M$ to problem (4.8), which satisfies*

$$\|Az\|_0 + \|s\|_{1/R} \leq c\|u - u^h\|_0.$$

Theorem 4.5. *Under the condition of Proposition 2.1 and Theorem 4.4, there exists a constant c independent of $h, \hat{h}$, such that the solution $(u^h, p^{\hat{h}})$ of (4.6)–(4.7) satisfies*

$$\|\nabla(u - u^h)\|_0 + \|p - p^{\hat{h}}\|_{0/R} \leq c(\hat{h} + |\ln h|^{1/2} \hat{H}^3),$$

and

$$\|u - u^h\|_0 \leq c(\hat{h}^2 + \hat{H}^3 \hat{h} + h\hat{H}^2).$$

Proof. Suppose that (w, π) is the projection of (u, p) on $X_h \times M_h$, and $w \in V_h$, then from (2.6), (2.14), (4.7), one finds

$$\begin{aligned} a(u^h - w, v) + d(v, \hat{p}^h - \pi) + d(u^h - w, q) &= \mathcal{B}((u^h - w, \hat{p}^h - \pi), (v, q)) \\ &= b(u, u, v) + b(u_H, u_H, v) - b(u^h, u_H, v) - b(u_H, u^h, v). \end{aligned}$$

We let $v = u^h - w$, by using (2.3) and Lemma 4.1, we obtain

$$\begin{aligned} v \|\nabla(u^h - w)\|_0^2 &= b(u, u, u^h - w) + b(u_H, u_H, u^h - w) - b(u^h, u_H, u^h - w) - b(u_H, u^h, u^h - w) \\ &= b(u - w, u, u^h - w) + b(w, u - w, u^h - w) - b(u^h - w, u_H, u^h - w) \\ &\quad - b(u_H - w, u^h - w, u_H - w) \\ &\leq C_b(\|\nabla(u - w)\|_0 \|\nabla u\|_0 \|\nabla(u^h - w)\|_0 + \|\nabla w\|_0 \|\nabla(u - w)\|_0 \|\nabla(u^h - w)\|_0) \\ &\quad + C_b \|\nabla(u^h - w)\|_0^2 \|\nabla u_H\|_0 + c |\ln h|^{1/2} \|\nabla(u_H - w)\|_0 \|u_H - w\|_0 \|\nabla(u^h - w)\|_0. \end{aligned}$$

So from Proposition 2.2 and Theorem 3.1, one finds

$$\|\nabla(u^h - w)\|_0 \leq c(\hat{h} + |\ln h|^{1/2} \hat{H}^3).$$

then from Proposition 2.1(ii) and Lemma 4.1, we obtain

$$\begin{aligned} \|\nabla(u^h - w)\|_0 + \|\hat{p}^h - \pi\|_{0/R} &\leq \sup_{\substack{(v,q) \in X_h \times M_h \\ \|(v,q)\| \leq 1}} |\mathcal{B}((u^h - w, \hat{p}^h - \pi), (v, q))| \\ &\leq c \sup_{\substack{(v,q) \in X_h \times M_h \\ \|(v,q)\| \leq 1}} |b(u, u, v) + b(u_H, u_H, v) - b(u^h, u_H, v) - b(u_H, u^h, v)| \\ &\leq c(\|\nabla u\|_0 + \|\nabla(u^h - w)\|_0 + \|\nabla w\|_0)(\|\nabla(u - w)\|_0 + \|\nabla(u^h - w)\|_0) \\ &\quad + c(\|\nabla(u^h - w)\|_0 \|\nabla(w - u_H)\|_0 \|\nabla(u^h - w)\|_0 \\ &\quad + c |\ln h|^{1/2} \|\nabla(w - u_H)\|_0 \|w - u_H\|_0) \\ &\leq c(\hat{h} + |\ln h|^{1/2} \hat{H}^3). \end{aligned}$$

So we have

$$\begin{aligned} \|\nabla(u - u^h)\|_0 + \|p - \hat{p}^h\|_{0/R} &\leq \|\nabla(u - w)\|_0 + \|\nabla(w - u^h)\|_0 + \|p - \pi\|_{0/R} + \|\pi - \hat{p}^h\|_{0/R} \\ &\leq c(\hat{h} + |\ln h|^{1/2} \hat{H}^3). \end{aligned}$$

Next we give the estimate of $\|u - u^h\|_0$.

Let $w = u - u^h$, $t = p - \hat{p}^h$ in (4.8), we obtain

$$|u - u^h|^2 = a(u - u^h, z) + b(u, u - u^h, z) + b(u - u^h, u^h, z) + (\nabla(p - \hat{p}^h), z) + (\nabla s, u - u^h).$$

From (2.6) and (4.7) we obtain

$$\begin{aligned} -a(u - u^h, v_h) - (\nabla(p - \hat{p}^h), v_h) - (\nabla q_h, u - u^h) &- b(u, u, v_h) + b(u^h, u^h, v_h) \\ &= b(u^h - u_H, u^h - u_H, v_h). \end{aligned}$$

From above two equalities we get

$$\begin{aligned} \|u - u^h\|_0^2 &= a(u - u^h, z - v_h) + (\nabla(p - \hat{p}^h), z - v_h) + (\nabla(s - q_h), u - u^h) \\ &\quad + b(u - u^h, u, z - v_h) - b(u^h, u - u^h, z - v_h) - b(u^h - u_H, u^h - u_H, v_h) \\ &\leq v \|\nabla(u - u^h)\|_0 \|\nabla(z - v_h)\|_0 + \|p - \hat{p}^h\|_{0/R} \|\nabla(z - v_h)\|_0 + \|s - q_h\|_{0/R} \|\nabla(u - u^h)\|_0 \\ &\quad + c \|\nabla(u - u^h)\|_0 \|\nabla u\|_0 \|\nabla(z - v_h)\|_0 + \|\nabla(u - u^h)\|_0 \|\nabla u^h\|_0 \|\nabla(z - v_h)\|_0 \\ &\quad + \|\nabla(u^h - u_H)\|_0^2 \|\nabla(z - v_h)\|_0 + c \|\nabla(u^h - u_H)\|_0 \|z\|_2 \|u^h - u_H\|_0. \end{aligned}$$

If we choose approximation (v_h, q_h) such that

$$\|\nabla(z - v_h)\|_0 \leq ch \|z\|_2 \leq ch \|u - u^h\|_0$$

and

$$\|s - q_h\|_0 \leq c\hat{h} \|s\|_{1/R} \leq c\hat{h} \|u - u^h\|_0,$$

because the asymptotic order of convergence of the fine grid solution $(u^h, \hat{p}^h)$ is at least as good as that of the coarse grid solution,

$$\|\nabla(u_H - u^h)\|_0 \leq \|\nabla(u - u^h)\|_0 + \|\nabla(u - u_H)\|_0 \leq C\hat{H}.$$

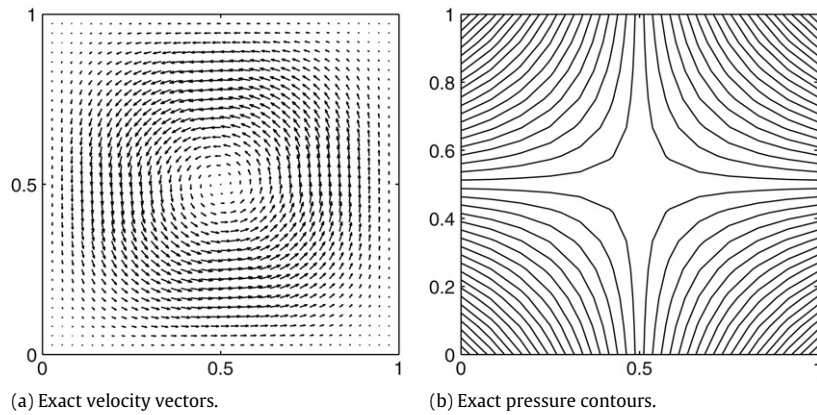


Fig. 1.

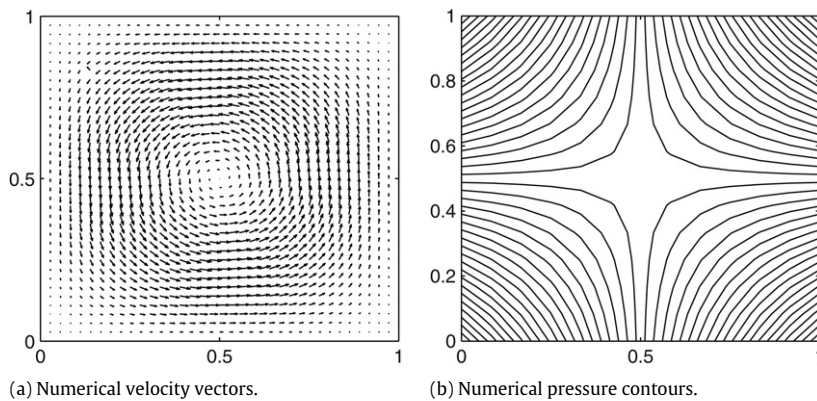


Fig. 2. GLM method.

So we have

$$\begin{aligned} \|u - u^h\|_0 &\leq c((h + \widehat{H}^3)h + \widehat{h}(h + \widehat{H}^3) + h\widehat{H}^2 + h\widehat{H}^3) \\ &\leq c(\widehat{h}^2 + \widehat{H}^3\widehat{h} + h\widehat{H}^2). \end{aligned}$$

5. Numerical examples

In this section we assess the performance of the GLM method described in Section 3 and the TGLM method described in Section 4. We consider a unit-square domain with a driven-cavity flow solution, which is a very popular problem in testing various numerical methods. In this paper, we set the exact solution as given by

$$\begin{aligned} u(x_1, x_2) &= (u_1(x_1, x_2), u_2(x_1, x_2)), \quad p(x_1, x_2) = 10(2x_1 - 1)(2x_2 - 1) \\ u_1(x_1, x_2) &= 10x_1^2(x_1 - 1)^2x_2(x_2 - 1)(2x_2 - 1), \\ u_2(x_1, x_2) &= -10x_1(x_1 - 1)(2x_1 - 1)x_2^2(x_2 - 1)^2 \end{aligned}$$

with $\nu = 0.05$ and f is determined by (2.1).

A practical problem description is shown in Fig. 1. In this paper, we use the $Q_1 - Q_1$ quadrilateral element, and the discretization parameters for velocity are smaller than that for pressure. When $h = 1/36$ and $h/\widehat{h} = 1/3$, the results of GLM, TGLM and Newton TGLM are presented graphically in Figs. 2–4, respectively.

In Table 1, we give the convergence accuracy of the GLM method. The results of the top table in Table 1 show that the relative error decreases with decreasing $h/\widehat{h}$, and this confirms the statement in the introduction. From the bottom table of Tables 1 and 2, we can conclude that the TGLM method is more efficient than the GLM method when they have the same order convergence rate. In Table 3, the convergence accuracy of the Newton TGLM method is provided.

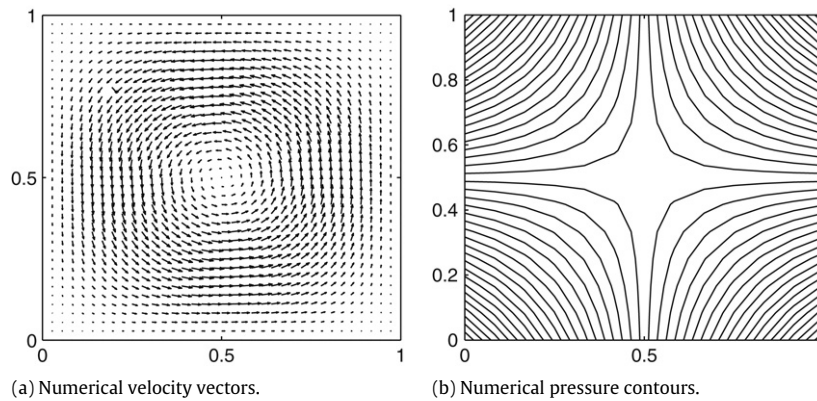


Fig. 3. TGLM method.

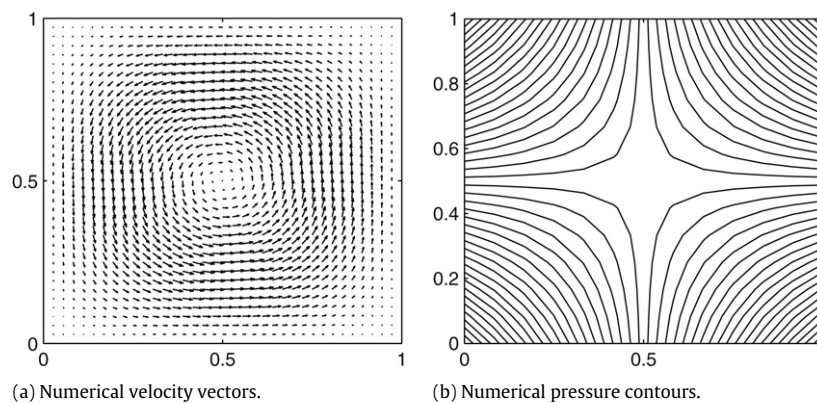


Fig. 4. Newton TGLM method.

Table 1

Numerical results of the GLM method.

h	$\hat{h}$	CPU(S)	$\frac{\ \nabla(u-u_h)\ _0}{\ \nabla u\ _0}$	$\frac{\ u-u_h\ _{L^\infty}}{\ u\ _{L^\infty}}$	$\frac{\ p-p_h\ _0}{\ p\ _0}$
1/18	1/6	67	0.0411777	0.0250273	0.000107423
1/24	1/6	329	0.040193	0.0189949	0.0000916813
1/36	1/6	3317	0.0386224	0.0164352	0.0000895317
1/18	1/6	67	0.0411777	0.0250273	0.000107423
1/27	1/9	937	0.03835323	0.0230255	0.000105059
1/36	1/12	5190	0.0381617	0.0225503	0.00010447

Table 2

Numerical results of the simple TGLM method.

H	$\hat{H}$	h	$\hat{h}$	CPU(S)	$\frac{\ \nabla(u-u^h)\ _0}{\ \nabla u\ _0}$	$\frac{\ u-u^h\ _{L^\infty}}{\ u\ _{L^\infty}}$	$\frac{\ p-p^h\ _0}{\ p\ _0}$
1/9	1/3	1/18	1/6	9	0.09121999	0.0148049	0.000182056
1/9	1/3	1/27	1/9	88	0.0855572	0.0139075	0.000177273
1/9	1/3	1/36	1/12	476	0.0631769	0.0118668	0.000168963
1/15	1/3	1/45	1/9	1650	0.0364177	0.0071132	0.000129472

Table 3

Numerical results of the Newton TGLM method.

H	$\hat{H}$	h	$\hat{h}$	CPU(S)	$\frac{\ \nabla(u-u^h)\ _0}{\ \nabla u\ _0}$	$\frac{\ u-u^h\ _{L^\infty}}{\ u\ _{L^\infty}}$	$\frac{\ p-p^h\ _0}{\ p\ _0}$
1/9	1/3	1/18	1/6	9	0.09122179	0.0156497	0.000171617
1/9	1/3	1/27	1/9	88	0.0828399	0.0126604	0.000161628
1/9	1/3	1/36	1/12	481	0.0636178	0.0108805	0.000159836
1/15	1/3	1/45	1/9	1622	0.0266339	0.0090577	0.000148731

References

- [1] I. Babuška, in: A.K. Aziz (Ed.), *The Mathematical Foundation of the Finite Element Method with Applications to Partial Differential Equations*, Academic Press, New York, 1972.
- [2] S. Falk Richard, An analysis of the finite element method using lagrange multipliers for the stationary Stokes equations, *Math. Comput.* 30 (1976) 241–249.
- [3] Ohannes A. Karakashian, On a Galerkin–Lagrange multiplier method for the Navier–Stokes equation, *SIAM J. Numer. Anal.* 19 (1982) 909–923.
- [4] J. Xu, A novel two-grid method for semilinear elliptic equations, *SIAM J. Sci. Comput.* 15 (1994) 231–237.
- [5] J. Xu, Two-grid finite element discretizations for nonlinear PDEs, *SIAM J. Numer. Anal.* 33 (1996) 1759–1777.
- [6] W. Layton, A two-level discretization method for the Navier–Stokes equations, *Comput. Math. Appl.* 26 (1993) 33–38.
- [7] W. Layton, W. Lenferink, Two-level picard and modified picard methods for the Navier–Stokes equations, *Appl. Math. Comput.* 69 (1995) 263–274.
- [8] W. Layton, L. Tobiska, A two-level method with backtracking for the Navier–Stokes equations, *SIAM J. Numer. Anal.* 35 (1998) 2035–2054.
- [9] V. Girault, J.L. Lions, Two-grid finite element scheme for the transient Navier–Stokes problem, *Math. Model. Numer. Anal.* 35 (2001) 945–980.
- [10] Yinnian He, Jian Li, Xiaozhong Yang, Two-level penalized finite element methods for the stationary Navier–Stokes equations, *Internat. J. Inform. Syst. Sci.* 2 (2006) 1–16.
- [11] Yinnian He, Kaitai Li, Two-level stabilized finite element methods for the steady Navier–Stokes problem, *Computing* 74 (2005) 337–351.
- [12] Maxim A. Olshanskii, Two-level method and some a priori estimates in unsteady Navier–Stokes calculations, *J. Comput. Appl. Math.* 104 (1999) 173–191.
- [13] Yinnian He, Two-level method based on finite element and Crank–Nicolson extrapolation for the time-dependent Navier–Stokes equations, *SIAM J. Numer. Anal.* 41 (2003) 1263–1285.
- [14] Yinnian He, A two-level finite Element Galerkin method for the nonstationary Navier–Stokes equations I: Spatial discretization, *J. Comput. Math.* 22 (2004) 21–32.
- [15] Yinnian He, Huanling Miao, Chunfeng Ren, A two-level finite element Galerkin method for the nonstationary Navier–Stokes Equations II: Time discretization, *J. Comput. Math.* 22 (2004) 33–54.
- [16] Yinnian He, K.M. Liu, Multi-level spectral Galerkin method for the Navier–Stokes Equations I: time discretization, *Adv. Comput. Math.* 25 (2006) 403–433.
- [17] Yinnian He, K.M. Liu, W.W. Sun, Multi-level spectral Galerkin method for the Navier–Stokes Equations I: spatial discretization, *Numer. Math.* 101 (2005) 501–522.
- [18] Yinnian He, K.M. Liu, A Multi-level Finite element method in space-time for the Navier–Stokes equations, *Numer. Methods Partial Differential Equations* 21 (2005) 1052–1078.
- [19] J.G. Heywood, R. Rannacher, Finite element approximation of the nonstationary Navier–Stokes problem I: Regularity of solutions and second-order error estimate for spatial discretization, *SIAM J. Numer. Anal.* 19 (1982) 275–311.
- [20] V. Girault, P.A. Raviart, *Finite Element Method for Navier–Stokes Equations: Theory and Algorithms*, Springer-Verlag, Berlin, Heidelberg, 1987.
- [21] Yinnian He, Huanling Miao, R.M.M. Mattheij, Z. Chen, Numerical analysis of a modified finite element nonlinear Galerkin method, *Numer. Math.* 97 (2004) 725–756.
- [22] C. Johnson, On computability and error control in CFD, *Internat. J. Numer. Methods Fluids* 20 (1995) 777–788.